

Ananya Penumudi

EMBEDDED & IOT ENGINEER

B.TECH CSE (IOT) · HYDERABAD

B.Tech CSE (IoT) student building **embedded systems, AI-powered devices and safety-critical software**. Industrial trainee at **BHEL Bengaluru** and author of a systems-engineering thesis on **KAVACH**, India's Automatic Train Protection system. CGPA 8.92.



PORTFOLIO

[ananyapenumudi-ops.github.io]

EMAIL

[ananyapenumudi@gmail.com]

LINKEDIN

[in/ananya-penumudi]

GITHUB

[github.com/ananyapenumudi-ops]

WORK EXPERIENCE

BHEL BENGALURU | 2026

Industrial Trainee · Jun – Jul

Trained at the Solar Business Division on manufacturing, production workflows, QA and system integration. Studied railway signalling and KAVACH through technical discussions and documentation.

→ **Certificate**

ALFAGO | 2026

Software Engineering Intern · Jan

Built tax-lot based profit-and-loss calculations and backend transaction logic for a Position Tracker covering transaction- and portfolio-level PnL.

→ **Certificate**

KARTGEN INFOTECH | 2025

Operations Intern · Jan – Apr

Remote internship in the Business Development Department: process coordination, stakeholder communication and day-to-day operations.

→ **Letter of recommendation**

LANGUAGES

English
Hindi
Telugu

SKILLS

- Embedded C/C++, Python, Java, JS
- Arduino, ESP32, SPI · I2C · PWM, BLE
- Computer vision & edge inference
- Functional safety, V&V, traceability
- React Three Fiber, FastAPI, MongoDB
- AWS, CI/CD, Tableau, Figma

EDUCATION

VNR VIGNANA JYOTHI INST. OF ENGG. & TECH. **2023 – now**
(B.Tech CSE (IoT) · CGPA 8.92)

SRI CHAITANYA JUNIOR COLLEGE **2021 – 2023**
(Intermediate · 97.4%)

BHARATIYA VIDYA BHAVAN'S PUBLIC SCHOOL **2021**
(CBSE · GPA 9.38)

FEATURED PROJECTS

ELEMENTIUM AI | 2026

3D virtual chemistry lab · React Three Fiber, FastAPI, Gemini

A 360° 3D lab with drag-and-drop apparatus and real-time titration. A Gemini-powered tutor and a rule-based engine catch unsafe or out-of-order steps, and the lab notebook fills itself in.

→ **GitHub**

KAVACH THESIS | 2026

Systems-engineering research · BHEL guidance

Analysis of the architecture, operating modes, communication framework and functional safety of India's Automatic Train Protection system, with a mode simulator and a digital twin.

→ **Thesis report** → **Simulator**

MORE PROJECTS

SMART AI WHEELCHAIR | 2026

IoT + edge AI

Vision-based obstacle detection with automatic braking, GPS geofencing, caregiver alerts and sensor fusion across camera, IMU and proximity sensors.

→ [Technical report](#)

SYNCOPE | 2025

Wearable IoT

A wearable that detects falls from motion-sensor data and alerts caregivers automatically.

→ [Documentation](#)

AURALYN | 2025 – NOW

Wellness wearable · in design

Designing an ESP32-S3 + BLE wearable that pairs biometric sensing with mood analysis, with apps for the wearer, therapist and family.

CERTIFICATIONS

DEVOPS WITH REAL-TIME PRACTICAL EXPOSURE

Brainovision · AICTE approved · Jul – Nov 2025

TABLEAU CERTIFIED DATA ANALYST TRAINING

Udemy · 59 hours · Jun 2024

→ [Certificate](#)

ROBOJAM ROBOTICS WORKSHOP

Convergence 2K23, VNRVJIET · Dec 2023

SECURE DATA WIPING | 2025

AI + cybersecurity · team project

Binary-overwrite data sanitisation with cryptographic verification of every wipe and AI-based anomaly detection.

→ [GitHub](#)

FOCUSFLOW | 2025

AI + web

An ADHD-friendly productivity platform with mood tracking, journaling, facial-expression analysis and gamified focus tools.

→ [GitHub](#)

EARLIER WEB PROJECTS | 2023 – 24

Ideavoid · Cheugy · Syndicate

A startup-idea validation platform, a stress and self-care app, and a gamified film-based learning platform.

→ [Cheugy on GitHub](#)

AWARDS & ACTIVITIES

- **2nd Prize, IDEATHON 2023**, VNRVJIET · **Certificate**
- **Smart India Hackathon 2025**: solution design, prototyping and technical documentation
- 24-hour Hackathon, Convergence 2025 · **Certificate**
- 24-hour Hackathon, DEMUX 2024, BVRIT · **Certificate**
- National Conclave 2025, CBIT · **Certificate**
- Member, Computer Society of India, VNRVJIET (2023 – now): organises workshops, coding events and competitions
- Member, Art of Living Cultural Club (2023 – now): events and volunteering